TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models

Daniel Kushnir KUSHNIR.DANIEL05@GMAIL.COM
Hao Wang (corresponding author) HOGUEWANG@GMAIL.COM
Rutgers University

Abstract

Time-series foundation models (TSFMs) are used zero-shot, but interpretability methods from language models do not transfer to them directly: tokens cover different spans of time in different architectures, real series offer no ground truth for concepts, and single internal features move a forecast so little that uncontrolled measurements yield confident false positives. We present TSFM-Lens, an open-source system that compares TSFMs on a measured common time axis, validates its instruments on a benchmark with known generative components and on a forecaster with planted concepts, defines concepts by their effect on the forecast relative to matched random interventions, and tests exploratory claims once on a sealed held-out split. Applied to seven TSFMs and confirmed on two independent corpora, it finds that all 21 model pairs share representational geometry above a shuffled-series null once each model's tokens are concatenated in time order; the usual window average understates TimesFM's similarity to four fine-token models by 0.30–0.43 CKA on held-out data. Different architectures also select the same input series for a concept: all registered transfer claims confirm on both private splits and on 800 raw real-world windows. Causal features act mainly on forecast level and seasonality, but fine-grained concepts are organized by model, and no concept spans more than three of the seven models. Activation prominence is a poor guide to causal importance (83.5% of prominent features have no held-out effect), and against random directions drawn from the activation distribution single features are only weakly specific (1.29 times chance). Every claim is reported with its evidence class and a reproduction pointer.

Keywords: mechanistic interpretability, time-series foundation models, sparse autoencoders, causal intervention, pre-registration

1 Introduction

Pretrained forecasting models such as TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024), Moirai (Woo et al., 2024), MOMENT (Goswami et al., 2024), Lag-Llama (Rasul et al., 2023) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. Their architectures differ sharply: decoder-only stacks over 32-step patches, encoder–decoders over quantized single steps, encoders with cross-series attention, decoders with sampled flow-matching heads. Leaderboards say which is more accurate. They do not say what these models represent, whether such different architectures represent the same things, or whether any internal structure we can name drives the forecast.

Mechanistic interpretability offers the instruments: probes (Alain and Bengio, 2017; Hewitt and Liang, 2019), representational similarity (Kornblith et al., 2019), activation patching (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024), lenses (Belrose et al.,

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2023) and sparse dictionaries (Bricken et al., 2023; Gao et al., 2025), packaged in libraries such as TransformerLens (Nanda and Bloom, 2022) and pyvene (Wu et al., 2024). Early work applies pieces of this toolkit to single TSFMs (Wiliński et al., 2025; Mishra, 2026; Oublal et al., 2026; Divo et al., 2026). Three properties of forecasting models prevent a direct transfer, and each produces silent errors rather than crashes.

(i) **Tokens are not comparable across models.** One token covers 1 time step in Chronos-T5, 16 in Chronos-2, 32 in TimesFM and 96 in Timer, so a cross-model comparison needs a verified mapping from tokens to time.

(ii) **There is no ground truth for concepts.** Real series carry no labels for trend, period, changepoints or noise process, so a probe that "finds seasonality" cannot be checked on them.

(iii) **Effects are small and easy to fake.** Removing one sparse feature typically moves a forecast by about a tenth of the context's standard deviation, less than the dictionary's own reconstruction error. At this scale a precision mismatch, an intervention the forecast head never reads, or a null measured on the wrong rows each produce a clean, confident and wrong result.

We present TSFM-Lens, a system built around these problems, and use it to ask four questions. **RQ1:** which forecast-relevant concepts do TSFMs learn? **RQ2:** are they shared across architectures, in the sense of responding to the same inputs, having the same effect, or both? **RQ3:** do they causally affect the forecast, and can their importance be read from activations? **RQ4:** where in depth do these computations happen? The system is *controlled*: a synthetic benchmark with exact generative ground truth and a forecaster with planted concepts give every instrument a case with a known answer. It is *causal*: a feature counts as a concept only if removing it changes the forecast more than removing a random direction of matched size on the same series. It is *confirmatory*: dev findings are frozen in a hash-pinned registry and tested once on a sealed private split.

Contributions.

C1. A comparative interpretability system (Section 3) with measured token-to-time alignment, a zero-code adapter path and a gated pipeline that turns a list of checkpoints into one report, together with a benchmark generator with leakage-audited tiers and sealed, bit-exactly regenerable splits.

C2. A causal concept methodology (Section 4) that defines concepts by their effect on the forecast and separates *same inputs* from *same effect on the same inputs*.

C3. Known-answer validation (Section 6): every null and agreement test is held to a criterion fixed in advance on a forecaster with planted concepts and decoys.

C4. A seven-model study with one-shot confirmation on two corpora (Sections 7 and 8), with series-level resampling, a multiplicity ledger and registered hypotheses (Section 5).

Main findings. We study TimesFM 2.5, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5-Base, Timer and Time-MoE, and re-test every registered claim on a second, three times larger corpus.

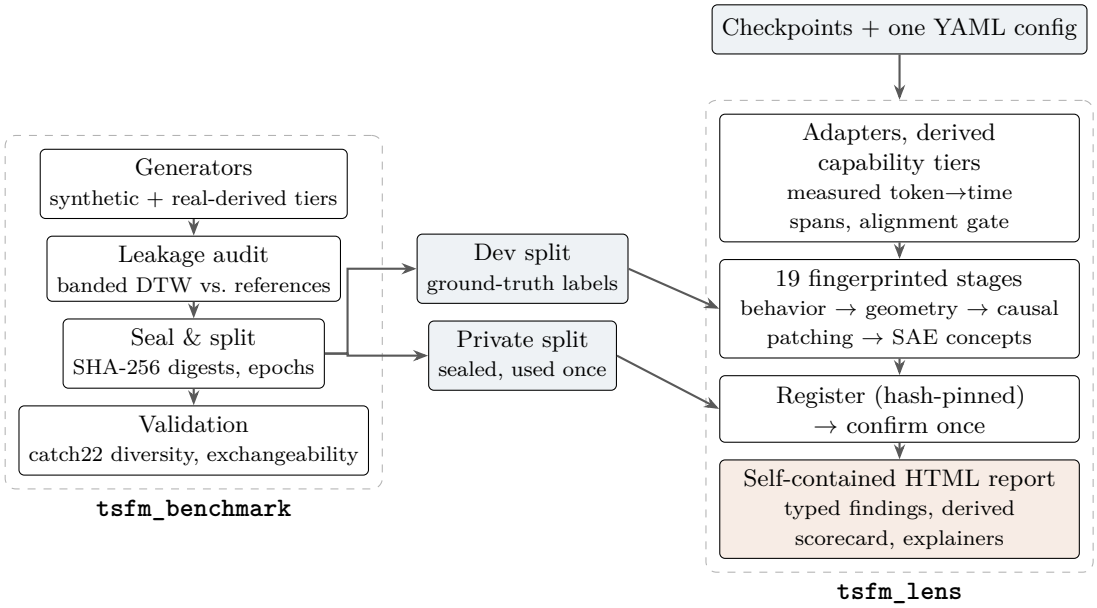


Figure 1: System overview. `tsfm_benchmark` produces leakage-audited, sealed corpora with exact ground truth; `tsfm_lens` turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by `confirm`, for claims registered in advance.

- **RQ1.** Sparse-dictionary features whose removal moves the forecast more than a random direction of the same size exist in six of the seven models. They act mostly on level and seasonality and form a graded continuum rather than discrete units. Against random directions drawn from the layer's own activation distribution they exceed chance only 1.29-fold.

- **RQ2.** All 21 model pairs share geometry above chance, confirmed on a fresh sealed split, provided windows concatenate their tokens; averaging them, the common practice, hides 0.30–0.43 of TimesFM's CKA with the fine-token models. Different architectures *select the same series* for a concept, held-out on both corpora and on raw real windows. Broad effect types recur in every model, but fine-grained concepts are organized by model: none spans more than three of seven.

- **RQ3.** Most prominent features have no held-out causal effect, and fewer than half of the registered single-feature causal claims replicate.

- **RQ4.** How TimesFM and Chronos-2 react to corruptions, layer by layer, replicates held-out for 6 of 9 corruptions on v2 and 4 of 9 on v3 (8 of 9 keep their sign).

Table 7 lists every headline result with its evidence class. Each result is an entry in the public findings ledger, `FINDINGS.md`, with exact numbers and a reproduction command; identifiers such as MN-29 refer to its entries and link to it.

2 Related Work

Time-series foundation models. TimesFM (Das et al., 2024) is a decoder-only model over 32-step patches. Chronos (Ansari et al., 2024) quantizes values and trains a T5 encoder–decoder (Raffel et al., 2020) with a sampled decoder; Chronos-Bolt replaces the decoder with a patch head, and Chronos-2 (Ansari et al., 2025) is an encoder with cross-series group attention. Sundial (Liu et al., 2025) samples from a flow-matching head (Lipman et al., 2023), Timer (Liu et al., 2024) uses 96-step tokens, Time-MoE (Shi et al., 2025) is a sparse mixture of experts over per-step tokens, and Lag-Llama (Rasul et al., 2023) reads lagged values. Most normalize each series internally (Kim et al., 2022). Synthetic data are common for pretraining and evaluation (Kang et al., 2020; Dooley et al., 2023); we use them to obtain ground truth for interpretation.

Interpreting time-series models. Saliency methods for time-series classifiers are unreliable across architectures (Ismail et al., 2020), and Kalnāre et al. (2025) apply patching and SAEs to classifiers. For TSFMs, Wiliński et al. (2025) study representational redundancy and concept steering, and four recent studies train SAEs on a single TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026). None compares architectures on a common time axis, validates its probes against generative ground truth, separates input agreement from effect agreement, or tests its findings on held-out data.

Validity of interpretability methods. Probes need control tasks (Hewitt and Liang, 2019; Belinkov, 2022); activation patching is sensitive to metric and corruption (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024); subspace interventions can create illusory mechanisms (Makelov et al., 2024), and a direction's apparent meaning can depend on the inspection data (Bolukbasi et al., 2021). Attributions can be insensitive to the model they explain (Adebayo et al., 2018), and attention weights are not explanations (Jain and Wallace, 2019). Sparse autoencoders are motivated by superposition (Elhage et al., 2022; Huben et al., 2024; Templeton et al., 2024), but their features vary across seeds (Paulo and Belrose, 2025); crosscoders share one dictionary across models (Lindsey et al., 2024). Causal abstraction (Geiger et al., 2021) formalizes when an intervention supports a mechanistic claim; our known-answer tests address the same concern empirically.

Comparing representations and research practice. Representational similarity (Raghu et al., 2017; Morcos et al., 2018; Kornblith et al., 2019; Kriegeskorte et al., 2008) and stitching (Lenc and Vedaldi, 2015; Bansal et al., 2021) underlie claims that networks converge on similar representations (Li et al., 2016; Huh et al., 2024); we use them only against shuffled-series nulls. Libraries such as TransformerLens, nnsight (Fiotto-Kaufman et al., 2025), pyvene, SAELens (Bloom et al., 2024) and Captum (Kokhlikyan et al., 2020) provide single-model primitives but not comparative designs, nulls, multiplicity control or confirmation (Table 1). From empirical science we adopt pre-registration (Nosek et al., 2018), protection against adaptive holdout reuse (Dwork et al., 2015) and analytic flexibility (Gelman and Loken, 2014; Simonsohn et al., 2020), leakage auditing (Kapoor and Narayanan, 2023) and series-level resampling (Efron and Tibshirani, 1993; Cameron et al., 2008).

Capability	*TransformerLens*	*nnsight / pyvene*	*SAELens*	*Captum*	*TSFM SAE studies*[†]	**TSFM-Lens**
Hooks, caching, activation patching	✓	✓	○	○	○	✓
Sparse dictionaries and feature ablation	○	○	✓	✗	✓	✓
Time-series models without per-model engineering	✗	○	✗	○	✗	✓
Measured token→time alignment across architectures	✗	✗	✗	✗	✗	✓
Benchmark with exact generative ground truth, leakage audit	✗	✗	✗	✗	○	✓
Random-direction nulls and reach controls for every ablation	✗	✗	✗	✗	✗	✓
Cross-model concept matching (inputs vs. effects)	✗	✗	✗	✗	✗	✓
Series-level resampling and multiplicity ledger	✗	✗	✗	✗	✗	✓
Registered, one-shot confirmation on sealed data	✗	✗	✗	✗	✗	✓
Typed evidence class and generated caveat per claim	✗	✗	✗	✗	✗	✓
One-command report designed for non-experts	✗	✗	○	✗	✗	✓

Table 1: Capabilities for comparative TSFM interpretability. ✓ provided; ○ possible with user code or partial; ✗ not provided. General libraries are single-model tools, complementary to TSFM-Lens, which uses the same primitives. [†]Mishra (2026); Oublal et al. (2026); Divo et al. (2026); Yıldırım (2026), judged from the papers.

3 The TSFM-Lens System

TSFM-Lens consists of two separately installable packages (Figure 1). `tsfm_benchmark` generates, audits, seals and validates corpora; `tsfm_lens` loads checkpoints through adapters, runs a gated pipeline of analysis stages and renders the report. Usage is described in Appendix A.

3.1 Design principles

Each principle is enforced in code as a gate, a refusal or a test, so that a measurement that cannot be trusted is withheld with a stated reason rather than reported.

P1. Ground truth by construction. Every probe is first validated on a case whose answer is known.

P2. Alignment is measured, not declared. Token spans are measured from impulse responses, and cross-model analysis happens only on a common time axis.

P3. Causal claims need a null, a reach control and a same-seed baseline. Every intervention is compared with a random intervention of matched size on the same series, and sampled models are seeded identically in compared calls.

P4. The series is the unit of inference. All intervals and p-values resample whole series.

P5. Explore, then confirm once. Registered dev claims are tested once on a sealed split.

P6. Every claim is typed. Each finding carries an evidence class and a generated caveat, and unsupported analyses are skipped with a rendered reason.

The evidence classes are BEHAVIORAL (input–output only), DESCRIPTIVE, ILLUSTRATIVE, GEOMETRIC (representational similarity), LINEAR-TRANSLATABLE (one representation linearly predicts another beyond an input baseline), CAUSAL WITHIN-MODEL (an intervention changes the forecast beyond a null), PREDICTIVE (internals forecast a model's error without being shown to cause it) and HELD-OUT CONFIRMED.

3.2 Controlled benchmark generation

Real data enter the benchmark only as leakage references, for realism calibration and as raw material for explicitly labeled generators. The *synthetic* tier touches no real data and has exact ground truth: an additive `parametric` generator (trend, multiple seasonalities, AR noise, changepoints, anomalies, random walk, intermittency, heteroskedasticity) and `random_parametric`, which draws a recipe from one of eight frozen archetypes. The *real-derived* tier (`mixture`, `block_bootstrap` (Künsch, 1989), `sequential_par`) inherits its source distribution and has attributional ground truth only. Every sample carries its tier label, so every analysis can be split by tier.

Every sample passes a two-stage leakage gate against the Monash archive (Godahewa et al., 2021) (a prefilter followed by banded DTW; Sakoe and Chiba, 1978) and a cross-split near-duplicate check. Dev and private splits use disjoint seed ranges, every sample is hashed so that loading a sealed split detects any change, and fresh private *epochs* are minted by striding the seed space. A validation package measures diversity in catch22 feature space (Lubba et al., 2019) by participation-ratio dimension (Gao et al., 2017) and nearest-neighbor tails, and tests dev/private exchangeability with an energy-distance test (Székely and Rizzo, 2013) and per-feature TOST equivalence (Schuirmann, 1987). Nine structured *corruptions* (noise, detrend, deseasonalize, frequency shift, level shift, spike, smooth, warp, dropout) drive the perturbation analysis, and *generator counterfactuals* move one generative parameter along a dose ladder with every other random draw held fixed.

3.3 Model adapters

An adapter exposes loading, a `predict` method returning point and quantile forecasts in the original scale, the underlying module, a forward pass and the time span of each token. Its *capability tier* is derived from the methods it implements: 0 (black box), 1 (observable hidden states with token spans), 2 (steerable, so patching is possible) and 3 (decomposable attention). Stages above a model's tier are skipped with the missing method named in the report, and a model whose tokens are measured not to be time-localized is routed to behavioral analysis. A zero-code adapter for Hugging Face checkpoints discovers layers and measures token spans; it reproduces the spans of all six hand-written adapters exactly and recovers Timer's 96-step tokens from measurement alone (MP-01). Two of the seven models studied here use it.

3.4 Time alignment across architectures

A 480-step context is 480 tokens in Chronos-T5 and Time-MoE, 30 in Chronos-2, Sundial and Chronos-Bolt, 15 in TimesFM and 5 in Timer, so raw token positions cannot be compared across models. We first measure each token's time span I_j by injecting an impulse at every time step; a token's span is the contiguous set of steps it responds to most, accepted only if it passes a peak-to-pedestal contrast gate (≥ 4) and a contiguity gate (≥ 0.95). We then place every model's tokens on one grid of windows $W_w = [wL, (w+1)L)$, $L = 32$ steps (Figure 2). A token shorter than a window contributes to it in time order, a token as long as a window maps to it one to one, and a token longer than a window (Timer's) is copied to every window it spans, never split. A window's representation is the concatenation of its tokens in time order,

$$\tilde{h}_w = \left[\, h_j \,\right]_{I_j \subseteq W_w}, \tag{1}$$

which loses nothing. Linear CKA needs only the row Gram matrix, and the Gram matrix of a concatenation is the sum of the per-position Gram matrices, $K = \sum_p X_p X_p^\top$, so the long vector is never formed. The common alternative, averaging a window's tokens, discards within-window order: in the first block of Chronos-T5-Base a window's slope decodes with $R^2 = 0.13$ from the window mean and 0.96 from finer sub-windows, while a TimesFM token keeps it. On held-out data the average understates TimesFM's similarity to the fine-token models by 0.30–0.43 CKA (Section 8.3; MN-38). Series-level vectors are means over the 15 windows.

What is compared across models. Alignment fixes the *rows* of two models' activations, not their coordinates: row (s, w) of every model describes series s over steps W_w, but widths range from 384 (Time-MoE) to 1,280 (TimesFM) and no axis of one model corresponds to an axis of another. Every cross-model measure is therefore defined on the shared rows. *Geometry* (CKA, and representational similarity analysis, RSA, on series summaries) compares how rows relate to one another within each model. *Stitching* fits a linear map from one model's rows to the other's and reports only its gain over a map from input features. *Concept transfer* compares which *series* a feature fires on, and *causal effects* are measured on the forecast. Patching never moves activations between models. Sparse dictionaries are fit to one model at a time and compared only through series identities and forecast effects, so their comparison does not depend on the window representation.

Two limits remain. The grid is no finer than the coarsest token: within a 96-step span Timer has one vector, so an alignment check can attribute an impulse to the right window at most one time in three, and hit rates are normalized by this attainable ceiling (MP-02). And $L = 32$ is a convention: re-pooling to 64 and 96 steps leaves concept transfer and the size of geometric similarity almost unchanged but moves the peak layer pair for 12 of 21 model pairs (SH-23), so peak locations are reported as window-dependent. During extraction an *alignment gate* injects an impulse into each window and checks, layer by layer, that the matching window responds most, with the amplitude calibrated per checkpoint and context length. Deep layers of Sundial, Time-MoE and Timer lose time localization, which the report shows next to window-level claims about those layers.

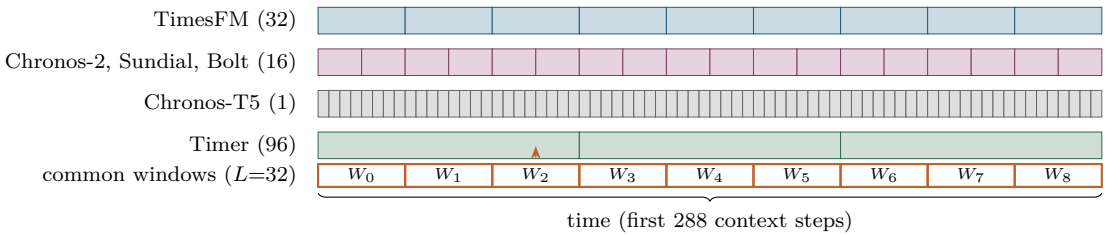


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5's one-step tokens are drawn in groups of four). Equation (1) places them on common 32-step windows: shorter tokens are concatenated in time order within a window, TimesFM's tokens map one to one, and each Timer token is copied to the three windows it spans.

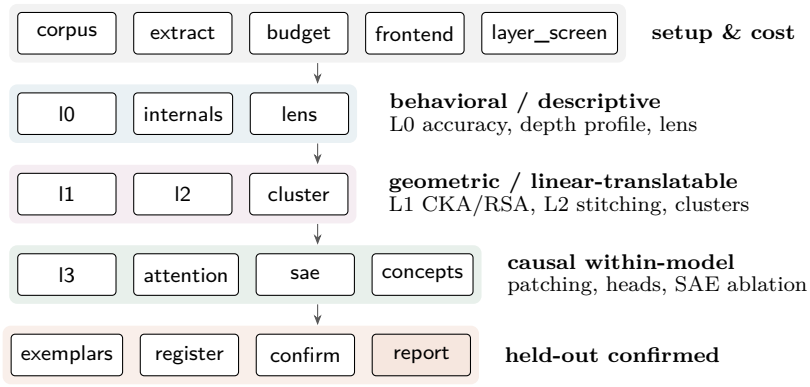


Figure 3: The 19-stage pipeline, grouped by the strongest evidence class each group can produce. Gates (tier, run shape, routing, configuration fingerprints) skip a stage per model with a rendered reason; only `confirm`, on the sealed private split, produces HELD-OUT CONFIRMED evidence.

3.5 The stage pipeline and the report

An analysis is a pipeline of 19 stages that write typed artifacts (Figure 3). Each stage declares the configuration keys it depends on, and these are fingerprinted, so a changed setting invalidates exactly the dependent cached artifacts. Three gates record their decisions, which the report names as skip reasons: the *tier* gate, the *run-shape* gate, which drops comparison stages for a single model, and the *routing* gate. Every intervention goes through a single patching primitive, which raises an error if the model silently splits a batch.

The deliverable is one self-contained HTML file, written so that a reader without interpretability training can follow it while an expert can audit every number; the seven-model example report is in `examples/panel7_v2`. Each stage opens with a card stating its question, method, what good and bad results look like and what it cannot show, and every figure carries an expandable explanation. The summary scorecard is computed from the artifacts by printed rules with no model names hard-coded: each row shows a value, the reference it is compared with (a null, floor, ceiling or baseline) and the rule that yields the verdict. Every claim is a typed record with its evidence class and a generated caveat.

4 Measurement Methods

Throughout, $x \in \mathbb{R}^T$ is a context, $y \in \mathbb{R}^H$ its continuation, $\hat{y} = f(x)$ a point forecast and $h^{(\ell)}$ the residual stream after block ℓ. The analyses are organized in levels: L0 (behavior), L1 (geometry), L2 (linear translatability) and L3 (perturbation and patching), followed by depth analysis and causal concepts.

4.1 Behavior, geometry and perturbation (L0–L3)

L0: behavior. Accuracy is the per-series MASE (Hyndman and Koehler, 2006), with the seasonal lag taken from ground truth and a floor on the naive denominator for intermittent series (MN-23). sMAPE, pinball loss (Gneiting and Raftery, 2007) and interval calibration are reported alongside. Model pairs are compared per generator family with a paired series bootstrap, Holm-corrected across all pairs and families (Holm, 1979), and differences are read against each model's repeat-run noise floor. Two further stages measure parameters, FLOPs, latency and the share of a forecast's FLOPs spent in captured blocks, and test what each model's input pipeline does before any layer runs (scale equivariance, handling of missing values, quantization).

L1 and L2: geometry. For aligned and centered window representations A and B over the same series, linear CKA (Kornblith et al., 2019), $\mathrm{CKA}(A, B) = \|B^\top A\|_F^2 / (\|A^\top A\|_F \|B^\top B\|_F)$, is computed for every layer pair of every model pair. A window is represented by its tokens concatenated in time order (Section 3.4), so CKA is computed from the row Gram matrices $K = \sum_p X_p X_p^\top$ summed over token positions p. Each value carries a series-level cluster bootstrap and a null that permutes whole series (1,000 permutations). L2 fits ridge maps between layers (Bansal et al., 2021) and reports only the *gain* over a baseline that predicts the target layer from input features, since much of any layer is linearly predictable from the input alone. Agreement between activation clusterings is measured by adjusted mutual information (Vinh et al., 2010).

L3: perturbation. For each corruption c, the sensitivity *fingerprint* of a model is the relative activation change $\|h^{(\ell)}(c(x)) - h^{(\ell)}(x)\| / \|h^{(\ell)}(x)\|$ as a function of layer, and its *restoration curve* is the fraction of forecast damage undone by patching clean activations of layer ℓ, whole or per window, into the corrupted pass. Two models are compared by the rank correlation of their fingerprints on a shared relative-depth axis, never by transplanting activations between models. Restoration is causal within a model; agreement between two models' fingerprints is correlational.

4.2 Depth

A skip lens decodes the forecast from each layer through the model's own head, and a tuned ridge lens (Belrose et al., 2023) fits a readout per layer. The depth axis is the fraction of the *whole* stack, including parts that are not captured: Chronos-T5's decoder is not captured, so its last captured encoder block sits at depth 0.478 (DE-04). The lens is withheld for a model whose head reads positions that patching never writes (DE-03).

4.3 Causal concepts

Sparse dictionaries. Each target layer receives a TopK sparse autoencoder (SAE) (Makhzani and Frey, 2013; Gao et al., 2025),

$$z(h) = \mathrm{TopK}\left(W_e(h - b_d) + b_e\right), \qquad \hat{h} = W_d\, z(h) + b_d, \tag{2}$$

with a searched dictionary size, an auxiliary loss that revives dead atoms and a minimum number of training steps (MP-03). Because single SAE seeds are not reproducible (Paulo and Belrose, 2025), three replicate dictionaries per target provide a within-model reproducibility ceiling (MP-04).

The ablation battery. Feature i is tested on the k series on which it fires most strongly. The forecast under the reconstruction with $z_i = 0$, written $\hat{h}_{-i}$, is compared with the forecast under the full reconstruction:

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \tag{3}$$

where g_c is one of nine forecast *channels*: trend, seasonal strength, spectral centroid, level, dispersion, near- and far-horizon shape distance, MASE and flatness. Using the reconstruction as the baseline keeps the SAE's reconstruction error, which moves the forecast more than a typical feature does (Section 8.4), in both arms. For a TopK dictionary zero is a feature's resting value, since at most 32 atoms are nonzero on any token, so the ablated state lies in the dictionary's normal range. The two shape channels are distances and are therefore treated as unsigned.

Random-direction nulls. The null asks whether removing the feature's direction matters more than removing the same amount of activation along another direction. Two variants are used.

- The *profile-matched* null (used for the v2 corpus) removes, on the same rows and at every token t, exactly the amount the feature removes, $|z_i(t)|\,\|w_i\|$, along each of 16 random *decoded* directions: a unit vector drawn isotropically in the dictionary's code space and mapped through the decoder. Only the direction differs from the real ablation. We call this null isotropic for short.
- The *covariance-matched* null (used for the v3 corpus) is identical except for the direction, which is a random Gaussian combination of the same rows' centered clean token activations. It is thus a draw from a Gaussian with the layer's sample activation covariance and lies in the subspace the model actually uses. It is the stricter of the two: on v3 it reduces the clearing cells from 4.92 times the empirical chance rate to 1.27 times (600 against 122.1 expected, then 170 against 133.5).

Feature i *clears* channel c if its mean $|\Delta_i^{(c)}|$ exceeds the 95th percentile $q_{95}^{(c)}$ of the pooled null, and its effect vector is $v_i^{(c)} = \overline{\Delta_i^{(c)}}/q_{95}^{(c)}$, the signed effect in units of its own null. Because a mean over rows is compared with a percentile of single rows, the chance rate of clearing is not 5%. It is estimated per cell by scoring each null draw against the others (leave-one-draw-out), and ranged from 0.3% to 4% per target in the case study.

Reach controls. Before any feature is scored, a *reach probe* requires that patching a layer's own activations back into it changes the forecast by exactly 0.0, and that patching another layer's activations changes it by a relative amount above 10^{-3}. A target that fails either check is withheld with its reason.

From features to concepts. Causal features are grouped at three granularities.

- *Per-target concepts* cluster one layer's effect vectors with k-means, choosing k by silhouette (Rousseeuw, 1987) among values whose smallest cluster has at least three members. Without this constraint the best silhouettes are one-outlier splits. A target with no admissible k is called *non-modular*.
- The *atlas* pools features from all models and layers and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a tight, shared causal-effect profile.
- *Families* are a coarser tiling for reading (average linkage at cosine 0.5), tested against a column-shuffle null and titled deterministically from their channels.

4.4 Two notions of sharing

Whether a concept is "shared" between two models is two separate questions.

Same inputs. *Concept transfer* asks whether some feature in another model separates a concept's top-k series from the rest, measured by AUC, against stratum-matched random subsets of series. It is run in both directions; a transfer is *reciprocal* if both directions clear, with Benjamini–Hochberg control (Benjamini and Hochberg, 1995) per ordered model pair and direction. Input agreement between two concept parts is the overlap of their top-k series (hypergeometric test), not a whole-series correlation, since two sparse features can correlate at $\rho = 0.80$ while sharing no top series (MN-19).

Same effect on the same inputs. For every reciprocal transfer, both sides are ablated on the same series set U. Each side must first clear its own null on U. The two sides then agree if their level concordance and shape cosine both exceed the floors given by ablating random feature sets matched to each side's activation. A statistic below both floors' 5th percentile is called *acts differently*; one that merely fails to exceed the 95th percentile is *no specific agreement*, since failing to beat a floor is not evidence of a difference (MN-18).

Evidence ladder and sharing classes. The report summarizes each atlas concept on a six-rung ladder: R1 same forecast effect (atlas membership), R2 same inputs, R3 reproducible across SAE seeds, R4 other dictionaries select the same inputs, R5 same causal effect on the same inputs, and R6 confirmed on private data (the report labels the rungs L1–L6). Each concept also receives a *sharing class*: *shared* (same effect and same inputs), *partially shared*, *convergent* (same effect reached on different inputs) or *single-model*. A concept part is flagged as provenance-driven when at least 80% of its top series come from one real-derived generator.

5 Statistical Protocol

Four mechanisms control how often a report with thousands of comparisons is fooled (Table 2).

The series is the resampling unit. Windows of a series share its trend, noise and scale, so every interval and p-value resamples whole series (Efron and Tibshirani, 1993; Cameron et al., 2008): paired bootstrap for accuracy, cluster bootstrap for CKA, validation-series intervals for stitching, paired cluster bootstrap for patching. Splits are by series and stratified.

A multiplicity ledger. Every test belongs to a named family, corrected by Holm for confirmatory families and by Benjamini–Hochberg for discovery-oriented atlas transfers. Resampled p-values are floored at $1/n_{\text{boot}}$ (Phipson and Smyth, 2010), so a Holm family of m tests cannot go below m/n_{boot}; the ledger computes this minimum per family and flags families that can never reach significance. The seven-model run has 123 families with 4,238 comparisons. The 8,705 feature-channel cells of the ablation battery are judged instead by their count against the empirical chance rate.

References measured in the run itself. Every effect is read against a reference measured in the same run: repeat-run noise floors, shuffled-series nulls, input-feature baselines, random-direction nulls, activation-matched floors or replicate-seed ceilings. Before the private split is opened, `confirm` simulates each claim's minimum detectable effect, for example a ΔMASE of 0.133 at $n = 290$ with 80% power.

Register, then confirm once. The `register` stage freezes hypotheses from dev results with the content hash of each source artifact, and hashes the registry. `confirm` refuses mismatched hashes, requires the private seal to verify, tests each claim once and refuses to overwrite its own artifact; a forced rerun is recorded as a `repeated_look`. A consumed split is replaced by a fresh epoch, checked for sample-hash overlap with every earlier split. This is pre-registration (Nosek et al., 2018) implemented in software, guarding against adaptive reuse of the holdout (Dwork et al., 2015; Gelman and Loken, 2014). Findings that were not registered cannot be confirmed after the fact.

6 Known-Answer Validation of the Causal Instruments

Every causal count in Section 8 rests on a null, and a null can be wrong without raising any error. Each instrument is therefore first tested on a forecaster whose answer is known.

The planted forecaster. TSFM-Lens ships a mock forecaster, `mock_planted`. One block writes known unit directions into the residual stream on the series that have a given property (thresholded readouts of the patch mean, slope, residual deviation, seasonal projections and a spike statistic, each firing on 10–30% of the corpus), and the forecast head applies a known effect on level, trend, seasonality or dispersion. Decoys look like concepts: *input-only* directions with zero causal effect and *sub-null* directions with a weak effect. A control layer after the planted block has no planted effect. For cross-model tests, a pair of architectures (widths 64 and 96, depths 5 and 7, rotated residual bases) shares planted concepts, plus a twin with the opposite effect.

Protocol. A scorer reads only the planted manifest and the run's artifacts. Pass criteria were fixed before any data were seen (Table 3): a per-cell false-positive rate of at most 0.10 on the control layer and a sensitivity of at least 0.5 for nulls, and a recall of at least 0.8 for

Claim type	Test statistic	Null / reference	Correction	Held-out
Model A more accurate (per family)	paired ΔMASE, series bootstrap	zero difference; repeat-run noise floor	Holm over all pairs × families	✓
Layers share geometry	linear CKA, cluster bootstrap	shuffled-series null at the same layer pair	per model pair	✓
Layer linearly translatable	ridge stitching gain	input-feature baseline (gain = 0)	per pair	–
Depth reaction agrees	Spearman ρ of fingerprints on shared depth	bootstrap CI of ρ	per corruption	✓
Layer restores forecast	restoration fraction	self-patch = 0, cross-patch ≠ 0	per model	–
Feature is causal	mean $\lvert\Delta^{(c)}\rvert$ on own rows	matched random-direction null, q_{95}	per channel (count vs. chance); Holm when registered	✓
Concept reproducible	best-match cosine across seeds	within-model replicate ceiling	–	–
Same inputs	top-k AUC in other model	stratum-matched random subsets	BH per ordered pair and leg; Holm when registered	✓
Same effect on same inputs	level concordance; shape cosine	activation-matched feature-set floors (q_{95} agree, q_{05} differ)	per test; Holm when registered	✓
Concepts are cross-model	atlas centroid cosine and member pairs	column-shuffle and model-label nulls	Holm when registered	✓
Internals predict failure	gain in log-MASE Spearman over a free baseline	zero gain	Holm when registered	✓

Table 2: Every claim type the case study reports, the statistic that tests it, the reference it must beat, its multiplicity correction and whether the `confirm` stage currently re-tests it on private data. Claim types without a held-out mark are reported as exploratory.

Instrument	Result on the planted forecaster	Pass criterion	Verdict
Profile-matched null (v2)	false-positive rate ≤0.003, sensitivity 0.70–1.00, all 5 seeds	FPR ≤0.10, sens. ≥0.5	pass
Covariance-matched null (v3)	sensitivity 0.916, false positives in 5 of 2,948 control cells	FPR ≤0.10, sens. ≥0.5	pass
Uniform batch-averaged null (reference)	false-positive rate 0.168–0.248 on every seed	FPR ≤0.10, sens. ≥0.5	fail
Gradient-attribution candidate ranking	mean top-20 recall 0.831 on unseen seeds	recall ≥0.8	pass
Shared-input agreement (R5)	false "same" 0 of 9 decoys; detects a shared one-direction concept in 0.4 of seeds, a 3–4-direction one in 0.0	–	specific, insensitive

Table 3: Known-answer tests of the causal instruments on the planted forecaster, with criteria fixed before each test. Ledger entries: MN-29 (nulls), MN-30 (agreement), MN-37 (attribution).

candidate ranking. Each is evaluated on five construction seeds at one effect size; seed 0 was used to tune the attribution rule and seeds 1–4 were held out.

Model	Checkpoint	Architecture	Token	Head	Tier	Params	FLOPs obs.
TimesFM 2.5	`google/timesfm-2.5-200m`	decoder-only	32	deterministic	3	231M	42.5%
Chronos-2	`amazon/chronos-2`	encoder (+group attn.)	16	deterministic	3	119M	97.5%
Sundial	`thuml/sundial-base-128m`	decoder + flow head	16	sampled	2	128M	21.3%
Chronos-Bolt	`amazon/chronos-bolt-small`	encoder + patch head	16	deterministic	2	48M	79.1%
Chronos-T5-Base	`amazon/chronos-t5-base`	encoder–decoder	1	sampled	3	201M	13.9%[a]
Timer	`thuml/timer-base-84m`	decoder-only	96	deterministic	2[b]	84M	99.8%
Time-MoE	`Maple728/TimeMoE-50M`	decoder-only, MoE	1	deterministic	2[b]	113M	50.0%

Table 4: The seven models. "Token" is time steps per token; "Tier" is the derived capability tier (Section 3.3); "FLOPs obs." is the share of a forecast's FLOPs spent in captured blocks at context 480, measured with PyTorch's FLOP counter. [a]Encoder only (DE-05). [b]Via the zero-code adapter. Adapters also exist for Lag-Llama, whose tokens read disjoint lags and therefore receive behavioral analysis only, and for other Chronos-T5 sizes (8M–709M).

Results. Both nulls used in the study pass on every seed (Table 3). The test discriminates: a null that removes one uniform, batch-averaged amount instead of the feature's own per-token profile fails on every seed, because a strong or dense atom clears it on size alone, and on a four-model development run it inflates the count of causal features from 110 to 293 (MN-29). On seed 0 the profile-matched null keeps every sub-null decoy from clearing, and raises the adjusted Rand index of the atlas against the planted concepts from 0.336 to 0.469 and sharing-class accuracy from 0.364 to 0.714.

The agreement rung R5 is specific but insensitive. It never calls a decoy "same", so its positive verdicts can be trusted, but it rarely detects a concept that two planted architectures truly share, and quadrupling the number of shared series does not change this (MN-30). An absent "same" verdict on real models is therefore not evidence of a difference. Known-answer tests establish what an instrument can detect and how often it is fooled in a world whose answer is known; they do not establish that its verdicts on real models are correct.

7 Experimental Setup

Models. We analyze seven TSFMs (Table 4): TimesFM 2.5 (Das et al., 2024), Chronos-2 (Ansari et al., 2025), Sundial (Liu et al., 2025), Chronos-Bolt and Chronos-T5-Base (Ansari et al., 2024), Timer (Liu et al., 2024) and Time-MoE (Shi et al., 2025). The panel covers decoder-only, encoder-only and encoder–decoder stacks, token widths from 1 to 96 steps, a mixture of experts, and deterministic and sampled heads.

Corpora. All claims are tested on two independently generated corpora, each with its own dev split, sealed private split and registry.

- **v2** (primary): 965 dev series in five generator families, 410 of them real-derived (Monash weather, ETT (Zhou et al., 2021), Chronos electricity and traffic), and a sealed private epoch of 971 series that shares no sample with any earlier split.
- **v3** (replication): 2,876 dev series, 785 of them real-derived from 15 sources (adding ERCOT load, M4 hourly, solar, taxi, KDD Cup air quality and pedestrian counts), and 1,024 private series drawn from a sealed epoch of 2,868. An *external* set of 800 raw windows from 12 GIFT-Eval datasets (Aksu et al., 2024), cut without any generator,

Model	median MASE	80% coverage	scale resid. (s.d.)	missing value	latency (ms)
Chronos-2	**1.2002**	0.757	0.0102	handled	23
TimesFM	1.2560	**0.779**	1.6×10^{-6}	propagates	39
Sundial	1.4131	0.360	0.0310	propagates	62
Chronos-T5-Base	1.4231	0.436	2.7×10^{-7}	handled	975
Chronos-Bolt	1.5487	0.732	0.0101	handled	15
Timer	1.9306	–	0.0262	propagates	5
Time-MoE	2.8642	–	0.0368	propagates	103
naive / seasonal naive	2.7593 / 1.9819				

Table 5: Behavior on the v2 dev split (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations, $c = 0.001$. Missing value: whether an injected NaN is handled or propagates silently. Latency: median time to forecast a batch of 8 series. Timer and Time-MoE output point forecasts only.

re-tests the registered transfer claims. A second sealed epoch (2,863 series) was opened once, for the registered window-geometry claims.

Dev and private splits are exchangeable on both corpora (energy-distance $p = 0.951$ and 0.995; 20 and 22 of 22 catch22 features equivalent by TOST, the rest inconclusive; MP-09). Contexts have 480 steps, a multiple of Timer's token, the horizon is 64 steps and the window 32 steps.

Protocol. On v2, SAEs are trained on 56 target layers (TimesFM 10, Chronos-2 9, Sundial 9, Time-MoE 9, Chronos-T5-Base 9, Timer 6, Chronos-Bolt 4), each with three replicate seeds, and causal features are scored against the profile-matched null. On v3, SAEs are trained on 50 targets and every causal claim is scored and registered against the covariance-matched null, with every setting fixed before the data existed. Bootstrap intervals use $n_{\text{boot}} = 5000$. The v2 registry holds 128 hypotheses, 82 replicable on the private split; the v3 registry holds 111, 63 replicable. Numbers are from the v2 run unless labeled v3; the v2 run took 21.1 hours on one RTX A5000, and its artifacts are in `examples/panel7_v2/run`.

How the models behave. Table 5 summarizes behavior on the v2 dev split. Chronos-2 is the most accurate. Five models beat the seasonal-naive forecast on 76.5–84.5% of series, Timer on 69.5% and Time-MoE on only 37.8%; Time-MoE is less accurate than the naive forecast overall, so its internal results describe a model that does not forecast these series well. The nominal 80% intervals of Sundial and Chronos-T5-Base cover only 36% and 44% of outcomes, and four models propagate an injected missing value silently. On the private split, Chronos-2 is more accurate than TimesFM on the registered `mixture` family (ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$) and, with a smaller effect, overall (-0.070 [-0.129, -0.008], $p = 0.025$, $n = 971$). On v3 the `mixture` result replicates (0.303 [0.176, 0.445], $p_{\text{Holm}} = 0.0002$).

Registered claim type	v2 (971 series)	v3 (1,024 series)	Section
Accuracy (Chronos-2 better on `mixture`)	1/1	1/1	7
Peak-CKA pair	1/1	1/1	8.3
Concept transfer (shared inputs)	20/20	20/20	8.3
same claims on 800 raw real windows	–	20/20	8.3
Atlas structure	2/2	2/2	8.3, 8.4
Single-model atlas concept	1/1	–	8.3
Cross-model causal (atlas concepts; agreement)	0/3; 0/6	none registrable	8.3
Single-feature causal effect	10/31	11/24	8.4
Perturbation-profile agreement	6/9	4/9	8.5
Failure prediction from internals	2/2	2/3	8.6
Confirmed / replicable (not testable)	43/82 (6)	41/63 (3)	
Window geometry, second sealed v3 split: 21 pairs above null; mean understates 4 pairs	–	21/21; 4/4	8.3

Table 6: Held-out confirmation on two independent corpora. Each cell is confirmed / tested; each claim type is one Holm family. The external raw-window set and the second v3 split (512 of its 2,863 series used) are not counted in the totals (CA-11, CA-12).

8 Results

We first summarize what replicated on held-out data and then answer the four research questions, each opening with its answer. Table 7 at the end lists every headline result with its evidence class.

8.1 Held-out confirmation at a glance

On v2, 43 of 82 replicable claims confirm, 33 do not and 6 cannot be tested; on v3, 41 of 63 confirm and 3 cannot be tested (Table 6). The pattern is the same on both corpora: claims about accuracy, geometry, shared inputs, atlas structure and failure prediction replicate, fewer than half of the single-feature causal claims do, and no claim that two models share a causal effect replicates.

8.2 RQ1: Which forecast-relevant concepts do TSFMs learn?

Answer. In six of the seven models, SAE features exist whose removal moves the forecast more than removing a random decoded direction of the same size. They act mostly on forecast level and seasonality and form a graded continuum rather than discrete units. Against random directions drawn from the activation distribution, few single features stand out.

Causal features beyond isotropic directions. Of the 56 target layers, 55 pass both reach controls; in Chronos-2's final block the head reads placeholder tokens that a context patch never reaches (DE-03). Across the 55 targets, 442 feature-channel cells clear the profile-matched null where the empirical chance rate predicts 178.75 (2.47 times chance). The ratio ranges from 1.64 (Chronos-T5-Base) to 5.70 (Chronos-Bolt) for six models; for Timer it is 1.02, although its layers do reach the forecast (PM-17). Target by target, 22 of 55 targets clear significantly more cells than chance (Benjamini–Hochberg at 0.05), at least

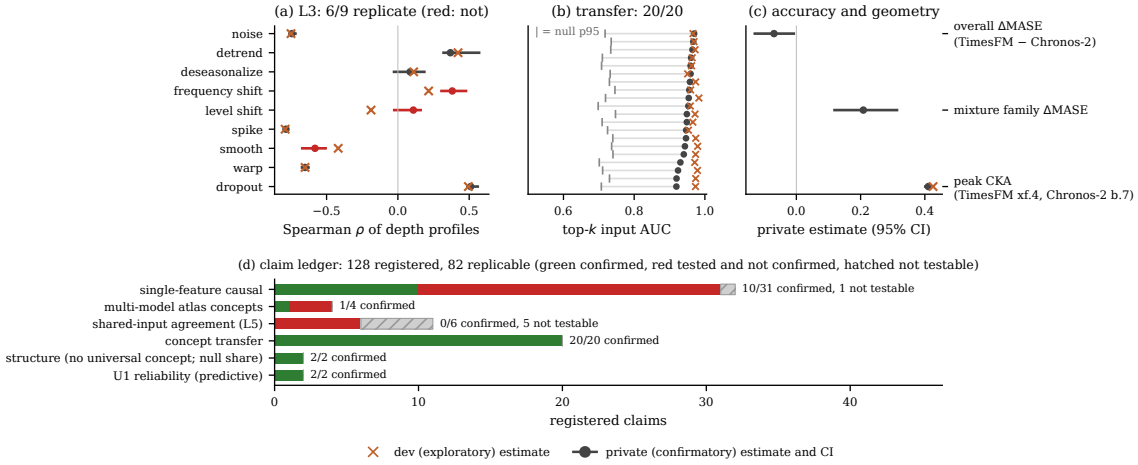


Figure 4: Claims re-tested on the sealed v2 private split (971 series; panel a uses 800 of them). (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2 (red: the dev value is not inside the private interval). (b) The 20 concept-transfer claims with private null 95th percentiles. (c) Accuracy and peak CKA; crosses are dev values, dots and bars private values with 95% series-bootstrap intervals. (d) Registered concept claims by type: confirmed, tested and not confirmed, and not testable (the one confirmed atlas claim is single-model). Source: `confirm/confirmation.json`.

two in every model except Timer. In all, 194 features clear at least one channel, against at most 144.95 expected if channels were independent.

The effects concern level and seasonality. Nine effect families tile the causal features (Figure 5). The two largest act on level: *level raisers* (45 features) and *level lowerers* (44), together 89 of the 192 assigned features. *Seasonality dampeners* (25) and *amplifiers* (19) follow; three mixed-effect families (14, 13 and 11), *trend boosters* (11) and *volatility dampeners* (10) are smaller.

The effects form a continuum. Within a layer, 50 of 56 targets are non-modular; the 10 per-target concepts that exist sit on 5 targets, 4 of them on Timer targets whose clear rate is at chance. The strict atlas assigns 35 of the 194 features to 11 concepts, more structure than a column-shuffle null produces (null 95th percentile 5.05 concepts; assigned fraction 0.180 against 0.088; $p = 0.005$) but covering few features. The families cover 192 of 194 features but do not beat the shuffle null in number ($p = 0.68$) or coverage ($p = 0.23$), and the silhouette is 0.25–0.32 for every cut from 6 to 16 families. The effect vectors are not low-dimensional (participation ratio 5.58 of 9 channels on v3; MN-07), so the absence of discrete families is not a lack of variation. The families are a vocabulary for reading the effects, not discovered units.

Against on-manifold directions, features stand out only modestly. The v3 run scored every feature against both nulls. Under the isotropic null, 238 of 967 scorable features clear at least one channel; under the covariance-matched null, 91 do. Scoring each null draw as a pseudo-feature on all nine channels at once, which keeps the dependence between

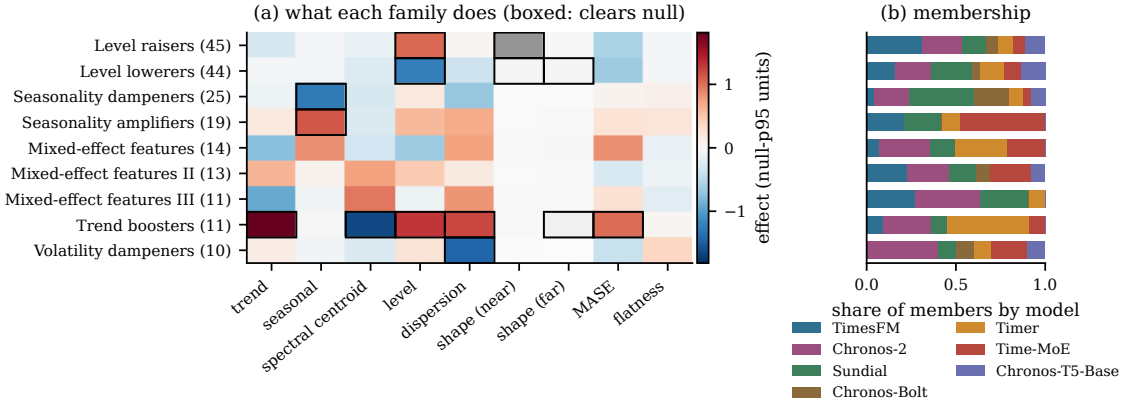


Figure 5: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature *does* (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

channels, gives expected counts of 67.0 and 70.4, so features exceed chance 3.55-fold against isotropic directions ($p = 4 \times 10^{-86}$) but only 1.29-fold against on-manifold ones (Poisson-binomial $p = 0.0045$). Even so, 11 of the 24 registered single-feature claims replicate on new series (Section 8.4).

Effects under on-manifold interventions. We re-ran the 91 v3 features that clear the on-manifold null under three further interventions, each with the null matched to its removal: setting the feature to its corpus mean (mean ablation), replacing it by its activation at the same token of another series on which it fires (resample ablation), and subtracting its decoder contribution from the model's own activation rather than from the reconstruction (clean ablation). Of the 91 features, 69 still clear under mean ablation, 58 of 90 scorable ones under resample ablation and 49 under clean ablation; 79 clear under at least one and 37 under all three. Mean and resample ablation keep the direction of the effect (median cosine to zero ablation 0.997 and 0.991), clean ablation less so (0.919). Survival under mean ablation depends on density: 56 of 62 sparse features survive but only 13 of 29 dense ones, for which the mean also adds the feature where it is silent. Of 91 sampled non-clearing features, 1, 2 and 8 clear under the three interventions (MP-10). Most causal features keep an effect under some on-manifold intervention, but which single features clear depends on the intervention, consistent with a graded continuum.

Alignment with generative factors is correlational. After dead-feature revival, 154 of 10,240 TimesFM features and 115 of 6,144 Chronos-T5-Base features correlate with a ground-truth generative field beyond a permutation null; the strongest is intermittency ($\rho = 0.881$; MP-06). The alignment does not extend to mediation: moving the generator parameter does not change a matched feature's forecast effect beyond chance (MN-17).

8.3 RQ2: Are concepts shared across architectures?

Answer. Whole-layer geometry is shared above chance by all 21 model pairs, held-out. Different architectures select the same series for a concept, held-out on both corpora and on raw real windows, and the broad effect types recur in every model. At a fine grain, however, concepts are organized by model, and the same causal effect on the same series is detected only between two Chronos-family models.

Geometry is shared above chance. With concatenated windows, peak CKA (the maximum over layer pairs) between TimesFM and each other model is 0.358–0.635 on v2 and 0.375–0.712 on v3 (512 series each), and all 21 model pairs exceed their shuffled-series null on both corpora ($p = 0.001$, 1000 permutations; null 95th percentiles 0.003–0.51, highest for pairs with Time-MoE). The layer pairs selected on v3 dev were registered and tested once on a fresh sealed split: all 21 exceed their nulls (Holm $p = 0.021$), with TimesFM/Chronos-2 at 0.707 [0.679, 0.734] against 0.712 on dev. At the same layers and series, averaging a window's tokens lowers TimesFM's CKA with Chronos-2, Chronos-T5-Base, Chronos-Bolt and Sundial by 0.322, 0.433, 0.301 and 0.303 (each 95% interval above 0.26; Holm $p = 0.0008$; MN-38). Pairs with Timer change little, because its 96-step token has no within-window order to match. Stitching agrees: concatenating the source window raises the mean gain over the input baseline at the 21 peak pairs from 0.498 to 0.554, most when TimesFM is predicted (from Chronos-T5-Base, 0.307 to 0.788), and keeps the ranking of pairs (Spearman 0.92).

Similarity has no single meaning. Different metrics rank the 21 pairs inconsistently (Kendall's $W = 0.25$, $p = 0.0095$, over seven metrics of the v2 run with mean-pooled CKA; Figure 6). TimesFM/Chronos-2 is the most similar pair by error agreement (0.948) but 15th by mean-pooled CKA, where Sundial/Chronos-Bolt ranks first (0.927); with concatenated windows TimesFM/Chronos-2 rises to 9th on v2 and 7th on v3. Geometry also does not reflect lineage: an untrained pair scores higher than a trained, related pair (MN-01). The report therefore never pools similarity metrics into one score and reads CKA only against shuffled-series nulls.

Models select the same inputs for a concept. On dev, 1,296 of 1,584 atlas transfer tests are reciprocal after Benjamini–Hochberg correction. Twenty transfer claims were registered from two source concepts, TimesFM layer 18 (14 claims) and Sundial layer 6 (6 claims). All 20 confirm on the 971 private series at the Holm floor for 20 claims at 2,000 null draws (0.0100), with AUCs of 0.919–0.970 against null 95th percentiles of 0.70–0.75 (Figure 4b), and the destinations span all seven models. Only 8 of the 20 best-matching private features are the ones selected on dev, so the confirmed claim is that the destination layer contains *some* feature selecting the same series. On v3, a new set of 20 claims (Sundial 14, Chronos-2 5, Time-MoE 1) confirms 20 of 20 on the private split and 20 of 20 on the 800 raw GIFT-Eval windows (AUC 0.757–0.991; SH-18). Re-pooled at 64 and 96 steps, the v3 reciprocal rate moves only from 0.827 to 0.801 and 0.799 (SH-23). This is evidence about inputs, not effects.

The broad effect types recur in every model. Level raisers, level lowerers and seasonality dampeners each have causal members in all seven models, and volatility dampeners

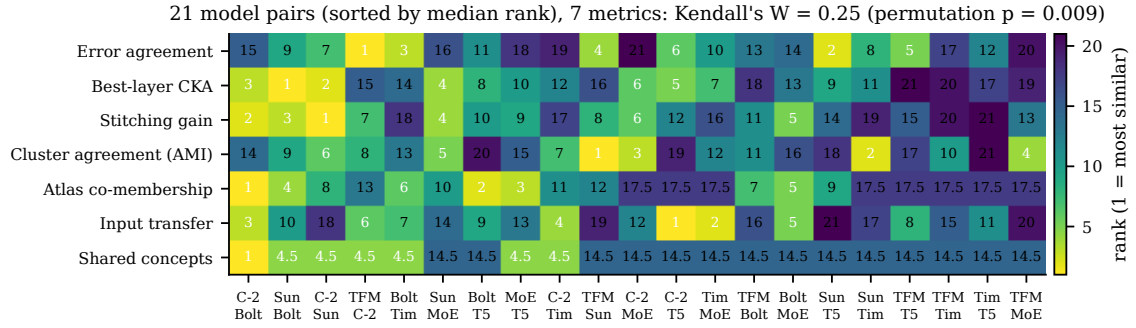


Figure 6: Rank of each model pair (columns, sorted by median rank; TFM TimesFM, C-2 Chronos-2, Sun Sundial, Bolt Chronos-Bolt, Tim Timer, MoE Time-MoE, T5 Chronos-T5-Base) under seven similarity metrics (rows); 1 is most similar. Source: `report/model_similarity.json`.

in six (Figure 5b). Membership reflects each feature's own measured effect, so it does not depend on whether the partition beats its shuffle null.

Fine-grained concepts are organized by model. Of the 11 atlas concepts, 9 have causal members from at least two models, but none spans more than 3 of the 7 models. The clusters are purer by model than random assignment makes them: 9 multi-model concepts against a null mean of 10.8 ($p_{\text{below}} = 0.025$), mean model purity 0.591 against 0.474 (SH-22). By sharing class, 2 concepts are shared, 2 partially shared, 5 convergent (the same effect reached on different series) and 2 single-model (Figure 7a). This holds held-out: no concept of the private atlas spans more than 3 of 7 models on v2 or more than 2 on v3. Each architecture builds its own variants of an effect, and several arrive at the same effect independently.

The same effect on the same series is detected only within the Chronos family. The shared-input test ablates both sides of a reciprocal transfer on the same series, each of which must clear its own null. Of 1,290 tests, 216 are scorable (Figure 7b): 7 show the same causal effect, 54 agree on level or shape alone, 133 show no specific agreement and 22 act differently. All 7 "same" verdicts are between Chronos-2 and Chronos-Bolt, six through one feature set in Chronos-Bolt's fourth block; 1,053 of the unscorable tests fail because one side does not clear its null on the 21–40 shared series. Because the test is specific but insensitive (Section 6), the 7 detections stand, while their absence elsewhere is not evidence that effects differ. Correspondingly, no registered cross-model causal claim confirms held-out: the three v2 multi-model atlas concepts exceed the null in centroid cosine (0.955–0.988 against 95th percentiles 0.854–0.907) but fail the registered conjunctive member-pair rule, and 0 of 6 testable shared-input agreement claims confirm.

8.4 RQ3: Do concepts causally affect the forecast?

Answer. Single features do causally shape the forecast, and some effects replicate on new series at the attainable significance floor. The effects are small, fewer than half replicate, and neither activation prominence nor attention mass identifies them.

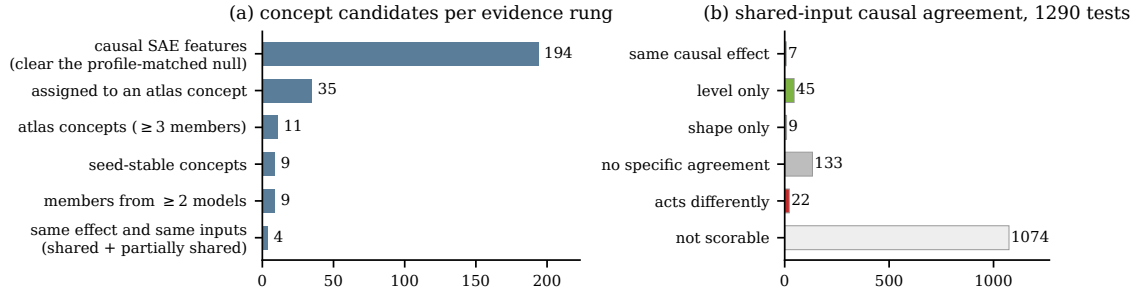


Figure 7: (a) Concept candidates at each rung (the first three bars are sequential; the last three are properties of the 11 atlas concepts). (b) Shared-input causal agreement over 1,290 tests drawn from the 1,296 reciprocal transfers. Source: `sae/concept_stage.json`.

Single-feature effects replicate partially. Of 31 testable single-feature causal claims registered from v2 dev, 10 confirm on the private split, each with $p = 0.001$ after an adaptive redraw to 1,000 null draws and $p_{\text{Holm}} = 0.031$, the floor for 31 tests. The confirmed share is highest for TimesFM (3 of 4) and lowest for Chronos-2 (1 of 7) and Chronos-T5-Base (0 of 4); by channel, near-horizon shape (4 of 8), MASE (3 of 8) and spectral centroid (2 of 4) confirm most often. Of the 21 claims not detected, 19 had not reached the 200-draw p floor on dev. On v3, under the stricter null, 11 of 24 claims confirm.

Activation prominence does not identify causal importance. Of the 927 most prominent features of the private atlas, 0.835 have no causal effect (one-sided lower bound 0.817; dev 0.810; $p_{\text{Holm}} = 0.0005$; MN-14); on v3 the share is 0.919 of 861. TSFM-Lens therefore ranks features by measured effect rather than by activation.

The effects are small. The median single-feature effect is 0.097 context standard deviations, 2.2 times smaller than the 0.214 by which substituting the SAE reconstruction moves the forecast (v3: 0.156 against 0.260, 1.7 times; MN-15). Among clearing cells, the median effect is 1.24 times the null's 95th percentile. The expressible forecast change is nonetheless concentrated: zeroing the five attribution-ranked atoms of a layer produces a median 0.74 share of it, more than activation-matched random sets of five atoms on 30 of 31 targets (D-42C).

Attention mass is not causal importance. The attention mass a head places on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ in TimesFM and -0.118 in Chronos-T5-Base; CA-04), consistent with Jain and Wallace (2019). A minimal seasonality circuit is one head in TimesFM but five non-additive heads in Chronos-T5-Base (CA-03).

8.5 RQ4: Where in depth, and how comparable is depth?

Answer. How TimesFM and Chronos-2 react to corruptions across depth replicates for 6 of 9 corruptions on v2 and 4 of 9 on v3, with 8 of 9 keeping their sign. Absolute depth locations are comparable only after accounting for captured compute.

Depth-resolved reaction profiles replicate. The rank correlation between the two models' per-layer corruption fingerprints replicates on the v2 private split for 6 of 9 corruptions (overall $\rho = 0.524$ [0.504, 0.544]; Figure 4a). They react at matching depths to detrending (private $\rho = 0.368$) and dropout (0.510), at opposite depths to noise (-0.743), spikes (-0.785) and warps (-0.651), and not systematically to deseasonalizing (0.086 [-0.027, 0.186]). Of the three that miss the registered rule (dev value inside the private interval), frequency shift (0.381 against 0.216 on dev) and smoothing (-0.581 against -0.418) are *stronger* on private data, while level shift changes sign (dev -0.188; private 0.109). On v3 level shift flips again (dev -0.164; private 0.252), the one depth agreement that is not stable across corpora.

Where restoration happens. In TimesFM, restoration by per-window patching concentrates in the window just before the horizon and grows with depth, from 0.427 at layer 0 to 0.612 at layer 8 and 0.631 at layer 16 for the noise corruption, while restoration in the first window falls from -0.191 to 0.000 (320 v2 series; DE-01).

Depth locations need qualification. Captured compute ranges from 13.9% (Chronos-T5-Base) to 99.8% (Timer) of a forecast's FLOPs (DE-05), and Chronos-T5's last encoder block sits at depth 0.478 on the whole-stack axis (DE-04); the report renders both next to every depth claim. No layer chosen by one of three layer selectors is robust to the choice of selector for any model (DE-08).

8.6 Can internals predict forecast failure?

For each model we predict a series' log MASE (Spearman) and a failure flag (AUROC) from free, output-only features (quantile width, flatness, catch22 features of the context), add the SAE-family activations at the last window, and measure the gain, cross-fitted with a series bootstrap on 941 dev series. The evidence class is PREDICTIVE. The free baseline is strong (Spearman 0.76–0.85), and for five models the gain's interval includes zero for both tasks. The gain is positive for the two least accurate models, Timer (+0.030 [0.019, 0.043]) and Time-MoE (+0.012 [0.005, 0.019]); both were registered and confirm when the frozen procedure is refit on the private split ($n = 935$): Timer 0.0446 [0.0292, 0.0605], $p_{\text{Holm}} = 0.002$, Time-MoE 0.011 [0.0032, 0.0194], $p_{\text{Holm}} = 0.012$, and again on v3 (MN-28). Internals do not help choose a model: routing each series to the model with the lowest predicted MASE gives 1.964 [1.837, 2.093] with internals against 1.948 [1.814, 2.070] without ($p = 0.258$), both worse than always using Chronos-2 (1.856).

RQ	Result (v2 unless stated)	Evidence class	Status	ID
1	442 feature-channel cells clear an isotropic null, 2.47× empirical chance; Timer 1.02×	causal within-model	exploratory	MP-07, PM-17
1	Against an on-manifold null, features exceed chance only modestly (v3: 91 vs. 70.4, 1.29×)	causal within-model	exploratory	CA-12
1	79 of 91 v3 causal features clear under mean, resample or clean ablation; 37 under all three	causal within-model	exploratory	MP-10
1	9 families cover 99% of features but do not beat a shuffle null ($p = 0.68$)	descriptive	exploratory	CA-10
2	All 21 pairs above null with concatenated windows; TimesFM/Chronos-2 0.712 dev, 0.707 fresh private; the window mean understates it by 0.30–0.43	geometric	held-out confirmed	SH-01, MN-38
2	7 similarity metrics agree only weakly ($W = 0.25$)	mixed	exploratory	SH-05/06
2	Concept transfer on inputs: 20/20 on v2 and v3, 20/20 on raw real windows	correlational	held-out confirmed (2 corpora)	SH-18
2	Level and seasonality-damping causal features in all 7 models	descriptive	exploratory	CA-10
2	No concept spans more than 3 of 7 models	descriptive	held-out confirmed (2 corpora)	SH-16
2	Same effect on same inputs: 7/216 scorable, all Chronos-2/Bolt; test specific (0/9 decoys)	causal on shared inputs	exploratory	SH-22, MN-30
2	Multi-model atlas concepts 0/3; shared-input agreement 0/6	causal, compared	not confirmed	CA-11
3	Single-feature causal effects: 10/31; v3 11/24	causal within-model	partly confirmed	CA-11/12
3	83.5% of prominent features causally null (927 held-out; v3 91.9%)	causal within-model	held-out confirmed	MN-14
3	Feature effect 2.2× below SAE reconstruction cost (v3 1.7×)	causal within-model	exploratory	MN-15
3	Attention mass uncorrelated with head causality	causal vs. descriptive	exploratory	CA-04
4	Depth-profile agreement 6/9 corruptions; v3 4/9; level shift flips on both	causal within-model	partly confirmed	CA-01
–	Internals predict failure for the two weakest models only	predictive	held-out confirmed (2 corpora)	MN-28
–	Chronos-2 more accurate than TimesFM on `mixture` (0.209; v3 0.303)	behavioral	held-out confirmed (2 corpora)	PM-05

Table 7: Headline results with evidence class and status. Each ID is an entry in `FINDINGS.md` with exact numbers, reproduction command and artifact path.

9 Discussion

Shared inputs, model-specific mechanisms. The most robust cross-model result is about inputs: models with different tokenizations, depths and training objectives single out the same series for a concept, on two corpora and on raw real windows. Whole-layer geometry is shared as well, most strongly at early, input-encoding layers, once windows are compared without averaging away their internal order. Yet at the grain of individual causal effects

concepts are organized by model, and agreement on the same series is detected only between two closely related Chronos models. One consistent reading is that the panel converges on *what* in a series matters (its level, its seasonality) while implementing the response differently. This is a weaker form of convergence than the representational convergence argued for in vision and language (Li et al., 2016; Huh et al., 2024), and our instruments locate the boundary: input agreement is measurable and replicates, whereas effect agreement across architecture families is below what ablation can resolve at these effect sizes (Section 6).

What the causal evidence supports. Single SAE features clearly beat isotropic random directions of the same size but only modestly beat directions drawn from the activation distribution, and which features clear depends on the intervention. The forecast-relevant information in a layer is spread over many directions, and removing any one typical direction moves the forecast by a comparable amount. Claims that a single feature "is" a forecasting concept should therefore be stated relative to an on-manifold null; against an off-manifold null a typical direction already looks special, the time-series counterpart of the illusions documented for subspace interventions (Makelov et al., 2024).

Implications for method. Three lessons generalize. First, activation-based selection is not a substitute for intervention: most prominent features have no causal effect, and attention mass does not predict head importance. Second, nulls must be validated on a known answer: a plausible uniform null inflated the count of causal features 2.7-fold. Third, a negative verdict needs a test of sensitivity: our agreement rung never produces a false "same" but rarely detects a planted shared concept, so its silence carries no information. Exploratory pipelines with thousands of comparisons (Gelman and Loken, 2014) benefit from the register-then-confirm discipline used here, which confirmed between half and two thirds of the registered claims (43 of 82 and 41 of 63). For practitioners, internal features add information about forecast failure only for the two least accurate models; the more immediate value of the system is diagnostic, from calibration and missing-value checks to instruments that refuse rather than return confident, wrong results.

10 Limitations

Models and alignment. The study covers seven transformer models with contiguous tokens; models whose tokens read disjoint lags, such as Lag-Llama, receive behavioral analysis only. With seven models, contrasts between models are confounded with architecture, size and training data, and Timer's features do not exceed chance, so its results carry no evidence about causal concepts. The analysis is univariate: Chronos-2's cross-series group attention is not measured (PM-13), and Chronos-T5's decoder and Sundial's flow head are not captured. A common time grid is no finer than the coarsest token, and peak layer locations depend on the window size (SH-23). Dictionaries are trained on window vectors, which is conservative for 1-step models: trained on its own tokens, Chronos-T5-Base's dictionary explains token variance far better and yields more causal features (MP-11).

Ground truth and external validity. Ground truth is exact only for the synthetic tier and encodes the generators' assumptions; 12 of 33 concept parts are dominated by one real-derived generator, and overlap with the models' pretraining data cannot be verified. Raw real benchmarks are a weaker basis for causal questions: real rows give TimesFM

representations of lower effective dimension (1.76 against 2.66) and more dead dictionary atoms (MN-20), and weakly predictable real-derived series receive flat forecasts in 67–100% of cases (BM-06; cf. Wan et al., 2026), leaving an ablation nothing to move. Real data therefore enter as real-derived series (410 of 965 v2 dev series), as dictionary-training rows and as an external check: the v3 transfer claims confirm on 800 raw real windows held out from every dev stage.

Scope of the confirmation. Held-out confirmation covers accuracy, geometry, perturbation agreement, transfer, single-feature causal effects, atlas structure and failure prediction on two corpora. The confirmed v2 transfers derive from two source concepts and the v3 transfers from three source models, so the transfer result rests on few concepts. Registered claims are selected from larger dev pools, which is legitimate because the private split referees them, but unregistered dev results remain exploratory. Per-series agreement of causal effects across architectures is not measurable by ablation at these effect sizes, so TSFM-Lens compares effects across models at the level of concept profiles.

Dictionaries and usability. Concept results depend on dictionary size, sparsity, revival and seed; the system measures this dependence but cannot remove it, and feature death tracks layer geometry (alive atoms scale with effective dimension with a log-log exponent of 1.02; MP-05). The controls catch only failures for which a known-answer case exists. Readability for newcomers is a design goal; no user study has measured comprehension.

11 Conclusion

Interpreting TSFMs is harder than it looks: tokens mean different things in different models, real series have no annotated ground truth, and causal effects are small enough that measurement artifacts dominate unless they are controlled. TSFM-Lens addresses these problems with a controlled benchmark, measured time alignment, null- and reach-gated causal concepts validated on a planted forecaster, and a register-then-confirm protocol, behind one command and a report written for newcomers. Across seven TSFMs and two corpora, different architectures share above-null representational geometry, a similarity that the common window average had hidden, and select the same inputs for a concept, while their causal machinery is organized by model; causal features act mainly on level and seasonality but are only weakly specific against on-manifold directions; activation prominence does not reveal causal importance; and depth-resolved reaction profiles replicate for most corruptions on the primary corpus and about half on the second. Each conclusion is stated with its evidence class, and each held-out claim was registered before the private data were opened.

Availability. Code (MIT license), configurations, the example run and the findings ledger are at `https://github.com/DanK10097110/TSFM-Interp`. Version 1.1.0, which this paper describes, is archived at doi:10.5281/zenodo.23267462, and both packages install from PyPI (`pip install tsfm-lens tsfm-benchmark`).

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Appendix A. Implementation and Usage

Repository layout. `tsfm_benchmark` generates, audits, seals and validates corpora; `tsfm_lens` holds models, extraction, analysis, sparse autoencoders, the report and the stage orchestrator. The folder `examples/panel7_v2` contains the report, configuration and every lightweight artifact of the v2 run, so the report and this paper's figures can be regenerated without a GPU.

A typical session. One installs the two packages, builds a benchmark and runs a configuration:

```
pip install -e . && pip install -e tsfm_lens
cd tsfm_benchmark && PYTHONPATH=. python3 example_runs/run_full.py \
       --config configs/example.yaml --out ./benchmark_out --references monash
cd ../tsfm_lens && python run.py --config configs/examples/medium_2model.yaml
```

The command-line interface adds a preflight check (`--doctor`), partial reruns (`--stages`, `--force`), checks for new checkpoints or library upgrades (`--check-alignment`, `--check-adapter`) and an environment diff (`--verify-provenance`). A CPU-only mock configuration runs every stage in minutes in continuous integration, and a worked example (`tsfm_lens/docs/worked_example.md`) walks through a report. In the v2 run, stages up to and including L3 take about 1.4 hours; most of the 21.1 hours are spent on SAE training and the concept battery.

Adding a model. A new checkpoint first tries the zero-code adapter. Of eight TSFM checkpoints probed, two (Timer and Time-MoE) load this way, and five that ship as standalone packages need a hand-written adapter (MN-25). For these, the repository provides a template (`tsfm_lens/tsfm_lens/models/TEMPLATE_adapter.py`), a conformance check (`--check-adapter`, which runs patch-identity and reach checks, the alignment gate, span discovery and capability checks) and a guide with a checklist of pitfalls (`tsfm_lens/docs/ADDING_A_MODEL.md`).

The example report. The seven-model report (`examples/panel7_v2/report.html`) contains 530 "What does this mean?" notes, one for every figure, and 240 typed findings (89 descriptive, 54 behavioral, 49 causal, 42 linear-translatable, 4 geometric and 2 illustrative). A glossary and a methods appendix are generated from the code.

Appendix B. Additional Validity Analyses

Benchmark quality. Every v2 sample passed the banded-DTW leakage gate (threshold 0.35) against Monash: 68 of 2,000 candidates were rejected, and the closest accepted sample

sits at distance 0.352. Between dev and private splits, tier and family composition are indistinguishable ($p = 0.94$ and $p = 0.9999$) and 6 of 937,015 cross-split pairs are near-duplicates (v3: 9 of 8,248,368). The dev split has effective catch22 dimension 5.37 of 22 (private 5.57) and a near-collision fraction of 0.052.

Statistical power. Each claim's power is stated. The 20 transfer claims all reach the Holm floor; the accuracy claim's minimum detectable ΔMASE is 0.133 against an observed 0.209; the held-out concatenated CKA of TimesFM and Chronos-2 is 0.707 [0.679, 0.734] against a null 95th percentile of 0.010. Single-feature causal claims are power-limited: each feature is tested on its 8 strongest-firing series, and 19 of the 21 that did not confirm never reached the dev p floor. The three times larger v3 corpus reproduces the held-out pattern (41 of 63 claims).

Robustness of the dictionaries. Each target has three replicate dictionaries, and 9 of 11 atlas concepts are stable across them. Held-out fraction of variance explained has medians of 0.647 (TimesFM) to 0.878 (Chronos-T5-Base). The ablation keeps the dictionary's error in both arms, so a measured effect is a real intervention whatever the fidelity, and fidelity does not predict causal clarity across v3 targets ($\rho = -0.326$, $n = 49$; MN-34). For the comparison of interventions in Section 8.2 the v3 dictionaries were retrained from the recorded configuration; training reproduced all 50 dictionaries exactly (identical loss histories) and the same 91 features clear.

Sampled models. Every compared pair of calls to Sundial or Chronos-T5 uses the same seed: unseeded, back-to-back Sundial calls differ by up to 0.50 MASE (PM-09). Repeat-run noise floors are reported per model (Sundial 0.14, Chronos-T5-Base 0.24 mean |ΔMASE|, the deterministic models 0).

Integrity of the confirmation. The v2 registry (128 hypotheses, sha256 recorded) was frozen before the private epoch was generated; the epoch shares no sample hash with any of seven earlier splits, and `confirm` ran once. Registered claims are cuts from larger dev pools (transfer 20 of 1,124 candidates, causal 32 of 113, agreement 11 of 25, atlas 4 of the 9 stable concepts). The v3 registry (111 hypotheses, sha256 `67b91269...`) was frozen in the same way, and the window-geometry claims were registered (sha256 `cda4f47a...`) before their epoch was extracted.

Appendix C. Statistical Details

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}}$ resampled means $\bar{d}^*_b$, the interval is the percentile interval and $p = \min\big(1, \max(\frac{1}{B}, 2\min\{\frac{1}{B}\#(\bar{d}^*_b \le 0), \frac{1}{B}\#(\bar{d}^*_b \ge 0)\})\big)$ (`analysis/stats.py`; floor per Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Holm: $\tilde{p}_{(i)} = \max_{j \le i} \min(1, (m - j + 1)p_{(j)})$ (Holm, 1979); with $p \ge 1/B$ the smallest attainable value is m/B, which the ledger reports per family, flagging $m/B > \alpha$. BH: $\tilde{q}_{(i)} = \min_{j \ge i} \min(1, \frac{m}{j} p_{(j)})$ (Benjamini and Hochberg, 1995).

Profile-matched ablation null. For feature i with rows R_i and $J = 16$ unit directions u_j, the null removes at every token t the amount $|z_i(t)|\,\|w_i\|\,u_j$; u_j is $W_d^\top r_j$ normalized, with r_j a random unit code. The covariance-matched null instead uses $u_j \propto \sum_t g_{jt}(x_t - \bar{x})$ with $g_{jt} \sim \mathcal{N}(0,1)$ over the chunk's clean token activations x_t, a draw from a Gaussian with their sample covariance. Channel c clears if $\frac{1}{|R_i|}\sum_{r\in R_i} |\Delta^{(c)}_{i,r}| > q_{0.95}\{|\Delta^{(c)}_{u_j,r}| : j \le J, r \in R_i\}$. A null with no spread is reported as degenerate and not scored.

Transfer. The statistic is $\max_f \mathrm{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k-sets, so the feature search is inside the null. The reverse leg scores the chosen feature's own top-k by the source activation. $p = (1 + \#\{\text{null} \ge \text{obs}\})/(1 + n)$; a registered claim uses $\max(p, p_{\text{rev}})$ (intersection–union), Holm across claims.

Shared-input agreement. On U, statistic (i) is the Spearman concordance of the two sides' signed level effects across series, and (ii) the cosine of their level-removed shape vectors, each channel in its own null units. Each side's floor ablates random feature sets matched to its activation on U and re-scores against the other side's measured effect; agreement exceeds both floors' $q_{0.95}$, "acts differently" falls below both $q_{0.05}$.

Minimum detectable effect. Before the private result is read, recentred per-series differences are shifted by a candidate effect and re-tested with the same bootstrap; bisection finds the smallest shift detected with power 0.8 (`analysis/power.py`).

Appendix D. Reproducibility

`examples/panel7_v2` holds the configuration, the report and every stage's JSON/parquet artifacts plus the confirmation (activations and SAE weights omitted for size); `paper/figures/make_figures.py --run DIR` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/panel7_v2.yaml` reproduces the run (`--stages report` re-renders it, `--verify-provenance` diffs environments). The v2 private split (epoch 3, 971 series) was opened once on 2026-10-02 and the v3 split (1,024 series) on 2026-10-10. One event is disclosed: the first v2 confirm attempt crashed while assembling the claim ledger, after the private batteries finished but before any statistic was written; no private number was read, the registry and every frozen specification were unchanged, and the artifact records `repeated_look: false`. Environment (`run_manifest.json`): Python 3.12.13, numpy 2.1.0, PyTorch 2.12.0 (CUDA 12.9), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7, scikit-learn 1.7.2; pins and reasons are in `DEPENDENCIES.md`. Tests: 2,647 (`tsfm_lens`) and 161 (`tsfm_benchmark`), run in continuous integration. Stage wall-clock on one RTX A5000: 75,900 s, of which `sae` 14,614 s and `concepts` 55,608 s.

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